The Insular Timing Hypothesis: Sensory Integration Failure, Feeding Difficulties, and the Gestational Fingerprint of Autism

Preprint – Not Yet Peer-Reviewed

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Date: 2026-03-05 **License:** CC-BY 4.0 International **Status:** Preprint – rejected by medRxiv (out of scope)

Abstract

Background: Between 46% and 89% of children with Autism Spectrum Disorder (ASD) experience feeding difficulties – a rate five times higher than neurotypical peers. These difficulties are among the earliest observable features, measurably diverging from typical trajectories by 15 months. Yet they remain clinically unexplained, categorised as behavioural rather than neurological. Simultaneously, the insular cortex – the brain's primary gustatory cortex, interoceptive hub, and multi-modal integration centre – undergoes its most dramatic structural transformation (opercularization, Von Economo neuron differentiation) during gestational weeks 20 – 28, precisely overlapping the pre-oligodendrocyte (preOL) vulnerability window established by Back et al. (2001). This convergence has not previously been examined.

Objective: To present the Insular Timing Hypothesis: that feeding difficulties, sensory oscillation, and the characteristic "spiky" sensory profile of autism arise from OPC maturation arrest affecting insular cortex white matter tracts during a defined gestational window. To demonstrate that the specific sensory modalities affected are determined by insult timing (TIMED selection), while the severity of expression at any given moment is modulated by physiological state (TEMPORAL expression) – producing the oscillation between hyper-sensitivity and hypo-sensitivity through a mechanism analogous to Uhthoff's phenomenon in multiple sclerosis. To map every major autistic symptom to a specific gestational week using the tract-specific myelination schedule, creating a forensic fingerprint of developmental timing.

Methods: Systematic integration of evidence from five domains: (1) gestational insular cortex development and gustatory/olfactory tract formation; (2) insular white matter abnormalities documented by DTI in ASD; (3) conduction biophysics of partially myelinated axons from the multiple sclerosis literature; (4) epidemiology of feeding difficulties in ASD and their relationship to sensory processing; (5) empirical feeding/sensory treatments that demonstrate efficacy without established mechanism. All claims graded using the evidence classification system established in Carpenter (2026a): Strong (>=3 independent PubMed-indexed replications), Moderate (1 – 2 published studies with plausible mechanism), or Speculative (mechanistically coherent but not yet empirically tested in this context).

Results: The overlap between the OPC vulnerability window (GW 23 – 32) and insular cortex development is extensive: insular opercularization (GW 20 – 28), gustatory thalamocortical connections (GW 24), Von Economo neuron differentiation (late gestation), and functional taste/smell activation (GW 28 – 30) all fall within or overlap the super-window. DTI studies confirm reduced integrity in both intrainsular tracts and the uncinate fasciculus in ASD, with intrainsular tract integrity directly correlating with tactile processing differences (Failla et al., 2017). The conduction biophysics of partially myelinated axons – established through decades of MS research – predict that circuits with a safety factor near unity will conduct unreliably: functioning under low load but failing under high-frequency stimulation, elevated temperature, or metabolic stress. This predicts two failure modes from a single mechanism: hypersensitivity (inhibitory circuit failure –> excitatory dominance) and hypo-sensitivity (complete conduction block –> signal dropout), with the mode determined by current physiological state. The sensory overload –> meltdown –> shutdown sequence maps to sequential conduction failure: inhibitory circuits (higher

metabolic demand) fail first (meltdown/disinhibition), followed by excitatory circuits (shutdown/block). A master timing overlay maps 15 major ASD symptom domains to specific gestational weeks using the tract-specific myelination schedule, demonstrating that the "spiky" autistic profile – proprioceptive and vestibular differences most affected, auditory and visual partially, language and social cognition emerging later – is precisely predicted by the myelination sequence. Ten empirical feeding and sensory interventions that demonstrate efficacy in ASD can be mechanistically explained as targeting taste receptor function, insular cortex processing, white matter connectivity, interoceptive regulation, vagal tone, or OPC/myelination support – with several targeting multiple mechanisms simultaneously.

Conclusions: The Insular Timing Hypothesis extends the gestational oxygen-timing alignment model to the sensory domain, proposing that the insular cortex is the single most vulnerable convergence point in the developing brain. Feeding difficulties in ASD are not behavioural – they are the clinical expression of multi-modal integration failure at a cortical hub that was under construction during the very weeks when oxygen disruption was occurring. The TIMED + TEMPORAL dual model is novel in its explicit formulation and integration of MS conduction biophysics with ASD sensory phenomenology. Five critical predictions are identified, each empirically tractable.

Keywords: insular cortex, sensory integration, feeding difficulties, autism, myelination, OPC maturation arrest, gestational timing, Uhthoff's phenomenon, gustatory development, Von Economo neurons, interoception, partial myelination, conduction failure

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Significance Statement

This paper proposes that the most accessible and visible feature of autism – difficulty with food – has a specific, identifiable, neurological cause rooted in gestational timing. By mapping the development of the insular cortex (the brain's taste, interoception, and emotional salience centre) onto the established OPC vulnerability window, we demonstrate that the same oxygen disruption that produces white matter injury also disrupts the hub responsible for integrating what food tastes like, feels like, and means. The "spiky" sensory profile of autism – extreme sensitivity in some modalities, minimal in others – is predicted by which nerve tracts were myelinating at the time of insult. The oscillation between hyper- and hypo-sensitivity follows from established conduction biophysics: partially myelinated circuits conduct unreliably, with reliability modulated by stress, fatigue, sleep, and hormonal state. This is the same mechanism that produces Uhthoff's phenomenon in multiple sclerosis – applied to autism for the first time. The model makes five empirically tractable predictions, explains why ten apparently unrelated feeding interventions all help, and provides a framework for forensic gestational timing based on an individual's symptom profile.

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1. Introduction

1.1 The Feeding Problem

Feeding difficulties affect between 46% and 89% of children with ASD (Sharp et al., 2013; Micai et al., 2023). ASD children are five times more likely to have feeding problems than neurotypical peers (Sharp et al., 2013; Campbell et al., 2024). These difficulties span multiple domains: food selectivity (~70%), food refusal (~40 – 60%), mealtime behavioural rigidity (~50 – 70%), and oral motor difficulties (~15 – 25%). They are among the earliest diverging features – measurably steeper trajectories from 15 months in prospective infant-sibling cohorts (Ashley et al., 2020) – and among the most distressing for families.

Yet feeding difficulties in autism are almost universally treated as a behavioural phenomenon. Intervention targets compliance, exposure, and desensitisation. The underlying question – *why* does this child experience food as aversive, overwhelming, or meaningless? – remains unanswered. No published framework connects ASD feeding difficulties to a specific neurodevelopmental mechanism with a de-

fined gestational origin.

1.2 The Insular Cortex – A Convergence Hub Under Construction

The insular cortex is unique in the developing brain. It is simultaneously:

- **Primary gustatory cortex** – receiving taste information via the ventral posteromedial nucleus (VPMpc) of the thalamus (thalamocortical connections established ~GW 24)
- **Interoceptive integration centre** – processing signals about internal body state (hunger, nausea, heartbeat, breathing)
- **Emotional salience processor** – generating the feeling of disgust, encoding the emotional meaning of sensory experience
- **Autonomic regulator** – modulating heart rate, gut motility, and arousal through Von Economo neurons (VENs)

The anterior insula sits at the intersection of what food *tastes* like, what it *feels* like in the body, what it *means* emotionally, and how the body *responds* autonomically. In a typically developing brain, these four streams are integrated seamlessly. In a brain where the insular cortex was under construction during the same weeks that oxygen delivery was compromised, the integration may fail – producing not "picky eating" but multi-modal sensory fragmentation.

1.3 The Developmental Overlap

The insular cortex undergoes its most dramatic structural transformation during the very weeks that define the OPC vulnerability window:

Gestational Week	Insular Development (Afif et al., 2007)	OPC/White Matter Status (Back et al., 2001)
13 – 17	First sulcus appears on insular surface	OPC specification phase
18 – 19	Periinsular sulci develop	Early preOL emergence
20 – 22	**Opercularization begins** – frontal, parietal, temporal opercula start covering insula	**PreOL population expanding**
24 – 26	**Posterior insula becomes covered**	**Peak preOL vulnerability – 72% death rate in ischaemia (Back et al., 2002)**
27 – 28	**Sylvian fissure closes; insula hidden**	**Super-window: 10/10 organ systems simultaneously vulnerable**
~28 – 30	**VEN differentiation begins** (Cauda et al., 2014)	**Ground zero for multi-organ injury**
~30	**Functional taste activation** via amniotic fluid (Paglia, 2019)	OPC window declining but still active

This overlap is not coincidental. The insular cortex is the *first* cortical region to begin sulcation/gyration (Afif et al., 2007), suggesting it is a central organising hub of early cortical development. Its myelination-dependent white matter connections – the uncinate fasciculus, extreme capsule, intrainsular tracts, and thalamocortical projections – are among the tracts most vulnerable to OPC maturation arrest.

1.4 What This Paper Proposes

This paper presents the **Insular Timing Hypothesis**: that the sensory integration difficulties, feeding problems, interoceptive differences, and oscillating sensory profiles characteristic of autism can be explained by OPC maturation arrest affecting insular cortex white matter tracts during a defined gestational window.

The model makes two linked claims:

1. **TIMED selection** – *which* sensory modalities are affected is determined by *which* white matter

tracts were myelinating at the time of the gestational insult. This produces the modality-specific "spiky" sensory profile.

2. **TEMPORAL expression** – *how severely* those modalities are affected at any given moment is modulated by current physiological state (stress, fatigue, sleep, hormones, temperature). This produces the oscillation between hyper-sensitivity and hypo-sensitivity.

The mechanism underlying temporal expression is analogous to Uhthoff's phenomenon in multiple sclerosis: partially myelinated circuits have a reduced safety factor for conduction, making them unreliable under physiological stress. This application of MS conduction biophysics to autism has not been previously proposed.

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2. Methods

2.1 Evidence Synthesis Approach

This synthesis integrates evidence from five domains:

1. **Developmental neurobiology** – gestational timelines for insular cortex formation, gustatory system maturation, olfactory system development, and white matter tract myelination
2. **Neuroimaging** – DTI and fMRI studies of insular white matter integrity and connectivity in ASD
3. **Conduction biophysics** – established electrophysiology of partially myelinated axons from the multiple sclerosis literature
4. **Clinical epidemiology** – feeding difficulty prevalence, sensory processing profiles, and pre-diagnostic markers in ASD
5. **Treatment evidence** – empirical feeding and sensory interventions with demonstrated efficacy but no established mechanism

2.2 Evidence Classification

All claims are graded using the system established in Carpenter (2026a):

Grade	Criteria	Symbol
Strong	>=3 independent PubMed-indexed studies with convergent findings; or meta-analytic confirmation	[STRONG]
Moderate	1 – 2 published studies with plausible mechanism and consistent direction	[MODERATE]
Speculative	Mechanistically coherent, supported by indirect evidence, but not yet directly tested	[SPECULATIVE]

2.3 Scope

This paper extends the gestational oxygen-timing alignment model (Carpenter, 2026a, 2026b) to the sensory domain. It assumes familiarity with the core model – OPC maturation arrest, HIF-1a pathway, the super-window concept, and the demand-spike/unmasking timeline. These are summarised where necessary but not re-derived.

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3. The Sensory Systems – Gestational Development Timelines

3.1 Gustatory System Development

Gestational Week	Developmental Milestone	Evidence
6 – 7	Nerve fibres approach lingual epithelium	Witt & Reutter, 1996 (PMID: 8955790)
8	Nerve fibres penetrate basal lamina – earliest taste bud primordium	Witt & Reutter, 1996
10 – 12	Type II and type III taste bud cells differentiated	Witt & Reutter, 1996
12 – 13	Synaptic density reaches maximum	Witt & Reutter, 1996
14 – 15	Taste pores form – structural prerequisite for transduction	Witt & Reutter, 1996
~24	Gustatory thalamocortical projections (VPMpc –> anterior insula) established	General thalamocortical timeline
~30	**Functional taste activation** – fetal taste receptors respond to amniotic fluid flavour variations from maternal diet	Paglia, 2019 (PMID: 31850766)

Key insight: Taste buds are structurally present from week 14 – 15, but functional taste perception via amniotic fluid does not occur until ~week 30. Before this, taste buds may serve non-gustatory paracrine functions (Witt & Reutter, 1996). The gap between structure and function is bridged by myelination of the gustatory thalamocortical pathway – a process occurring within the OPC vulnerability window.

Evidence strength: [MODERATE] (Moderate – primary developmental anatomy literature)

3.2 Olfactory System Development

Gestational Week	Developmental Milestone	Evidence
~5	Nasal placodes form; olfactory pit invagination	Sarnat & Flores-Sarnat, 2019 (PMID: 31604554)
~6 – 7	Olfactory axons project toward telencephalon *before* olfactory bulbs form	Sarnat & Flores-Sarnat, 2019
14	Histologic lamination of olfactory bulb	Sarnat & Flores-Sarnat, 2019
20	Fetal olfactory bulb tissue available for immunohistochemistry	Liu et al., 2010 (PMID: 19899113)
28 – 30	**Functional onset** – discrimination of odorous molecules in amniotic fluid	Sarnat, 2017 (PMID: 28424010)
>=30	Olfactory bulbs reliably demonstrable by MRI	Sarnat & Flores-Sarnat, 2019
At term	**Olfactory maturation remains incomplete** – myelination occurs postnatally	Sarnat & Flores-Sarnat, 2019

Unique feature: The olfactory system is the only sensory system without a thalamic relay. The olfactory bulb and anterior olfactory nucleus serve as intrinsic "thalamic equivalents." This means olfactory signal processing is structurally distinct from all other sensory modalities – and may therefore show a different vulnerability pattern.

Key principle from Sarnat (2017): "Immaturity does not signify nonfunction." The olfactory system functions before it is structurally mature. This has implications for understanding pre-term sensory ex-

perience and its role in programming sensory preferences.

Evidence strength: [STRONG] (Strong – dedicated developmental olfactory literature)

3.3 Insular Cortex Structural Development

Stage	Gestational Weeks	Key Development	Evidence
1	**13 – 17**	First sulcus appears – earliest macroscopic cortical feature	Afif et al., 2007 (PMID: 17962979)
2	**18 – 19**	Periinsular sulci develop (superior and inferior limiting sulci)	Afif et al., 2007
3	**20 – 22**	Central sulci form; **opercularization begins**	Afif et al., 2007
4	**24 – 26**	Posterior insula becomes covered by opercula	Afif et al., 2007
5	**27 – 28**	Sylvian fissure closes; insular cortex largely hidden	Afif et al., 2007

Von Economo Neurons (VENs): First differentiable at late gestation, increasing in number through infancy (Cauda et al., 2014; PMID: 24672457). VENs are found exclusively in the anterior insula and anterior cingulate cortex. They are critical for interoception, social cognition, and self-awareness. Their differentiation during the OPC vulnerability window makes them directly vulnerable to gestational oxygen disruption.

The insula as developmental organiser: The insular cortex is the first cortical region to begin sulcation/gyration (Afif et al., 2007), with right hemisphere development slightly preceding left. This suggests it serves an organising role in cortical development – making its disruption potentially consequential for broader cortical circuit formation.

Evidence strength: [STRONG] (Strong – definitive developmental study with 21 fetal brains)

3.4 Insular White Matter Tracts

The insular cortex is connected by five major white matter tract systems, each with its own myelination timeline:

Tract	Insular Relationship	Myelination Period	Functional Domain	ASD Evidence
Uncinate fasciculus (UF)	Distinct insular segment (isthmus); connects amygdala –> orbitofrontal cortex	5 – 30 years (peak FA ~30; Lebel et al., 2008)	Social reward, emotional regulation, word retrieval	Left UF FA reduction confirmed by meta-analysis (Aoki et al., 2013; PMID: 23876131)
Extreme capsule (EmC)	Located directly between claustrum and insula	Postnatal – poorly characterised	Ventral language stream, interoceptive signals	ASD-specific diffusivity differences (Kilroy et al., 2022; PMID: 36376319)

Tract	Insular Relationship	Myelination Period	Functional Domain	ASD Evidence
Intrainsul tract	Posterior insula –> anterior insula	Postnatal – poorly characterised	Connects somatosensory (posterior) to emotional salience (anterior)	Reduced integrity ((up)MD) in ASD; directly correlated with tactile processing (Failla et al., 2017; PMID: 28630661)
Arcuate fascicu-lus (AF)	Passes directly beneath insular cortex	3 – 15 years (peak ~12 – 15; Friederici, 2017)	Complex syntax, language	**The key ASD language tract** – unmyelinated at birth
Thalamoc projec-tions	VPMpc –> anterior insula (gustatory); central amygdala –> insula	~GW 24 onward	Taste, emotional salience relay	Not directly studied in ASD DTI

Critical finding – the uncinate fasciculus: The UF has the longest developmental period of any white matter tract, continuing to myelinate until approximately age 30 (Lebel et al., 2008; PMID: 18295509). It is therefore vulnerable to disruption for the longest period. Preterm-born 12-year-olds show UF abnormalities correlating with reduced verbal and full-scale IQ (Constable et al., 2008; PMID: 18245422). Children who experienced socioemotional deprivation show reduced UF fractional anisotropy (Eluvathingal et al., 2006; PMID: 16740852). In ASD, left UF FA reduction is confirmed by meta-analysis across 387 participants (Aoki et al., 2013).

Evidence strength: [STRONG] (Strong for UF; Moderate for intrainsular tract; Emerging for EmC)

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4. The Timing Overlay – GW 23 – 32 and the Insular Convergence

4.1 The Super-Window Alignment

At the established super-window (GW 24 – 32), the insular cortex is undergoing its most critical developmental transformations simultaneously with peak OPC vulnerability:

System	Developmental Event During GW 23 – 32	Evidence
Insular cortex (structure)	Stages 3 – 5: opercularization, posterior covering, Sylvian fissure closure	Afif et al., 2007
Insular cortex (VENs)	Von Economo neuron differentiation begins	Cauda et al., 2014
Gustatory pathway	Thalamocortical connections form (~GW 24); functional taste activation (~GW 30)	Paglia, 2019
Olfactory system	Functional onset (~GW 28 – 30); olfactory bulbs visible on MRI from GW 30	Sarnat, 2017
Insular white matter (UF)	Uncinate fasciculus development – insular segment forming	Lebel et al., 2008
Insular white matter (EmC)	Extreme capsule between claustrum and insula – formation phase	Kilroy et al., 2022
White matter (preOLs)	Peak vulnerability – 72 +/- 6% death rate in ischaemia	Back et al., 2002

System	Developmental Event During GW 23 – 32	Evidence
Cortical subplate	Peak functional importance; transient population (ZERO regeneration)	Carpenter, 2026b

4.2 What Oxygen Disruption During This Window Would Produce

If placental insufficiency disrupts oxygen delivery during GW 23 – 32, the insular cortex would sustain a convergent four-level injury:

Level	Injury	Downstream Consequence
1. Structural	Opercularization disrupted –> abnormal insular surface morphology	Altered cortical folding –> reduced surface area for integration
2. Cellular	VEN differentiation arrested –> reduced VEN count	Impaired interoception, social cognition, autonomic regulation
3. Connectional	OPC maturation arrest in UF, EmC, intrainsular tracts –> partial myelination	Unreliable signal transmission between posterior (somatosensory) and anterior (emotional/gustatory) insula
4. Functional	Gustatory and olfactory thalamocortical circuits incompletely myelinated	Taste/smell signals arrive with unpredictable timing –> integration failure

The clinical expression: A child whose insular cortex sustained this convergent injury would experience food not as a unified percept (taste + texture + temperature + smell + emotional meaning + body-state response) but as a fragmented set of overwhelming, unreliable, or absent signals. This is not "picky eating." It is multi-modal sensory integration failure at the hub responsible for binding food experience into a coherent whole.

Evidence strength: [SPECULATIVE] (Speculative – individual components are well-evidenced but their convergent effect at the insular cortex has not been tested as a unified hypothesis)

4.3 Consistency with the Multi-Organ Crossover Model

This insular timing hypothesis adds a previously unmapped system to the established super-window framework (Carpenter, 2026b). The timing overlay is consistent: the insular cortex vulnerability window (GW 20 – 28 for structure; GW 24 – 30 for gustatory/olfactory function) falls within the established 23 – 36 week super-window during which 9 – 10 organ systems are simultaneously vulnerable. The insular cortex can now be added as an eleventh system:

#	System	Window Start	Peak	Window Close	Regeneration
1 – 10	*As established in Carpenter (2026b)*	–	–	–	–

#	System	Window Start	Peak	Window Close	Regeneration
11	**Insular cortex (gusta-tory/interocep**	**20**	**24 – 28**	**30**	**VENs: zero. White matter: OPC-dependent**

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5. The TIMED + TEMPORAL Dual Model of Sensory Oscillation

5.1 The Clinical Observation: Hyper- and Hypo-Sensitivity Co-Occur

The co-occurrence of hyper-sensitivity AND hypo-sensitivity within the same autistic individual is the dominant sensory pattern in ASD – more characteristic than either direction alone:

Study	Design	Key Finding	PMID
Ben-Sasson et al., 2009	Meta-analysis, 14 studies, N>1,000	Hyper- and hypo-responsivity co-occur MORE in ASD than typical development	18615687
Ausderau et al., 2014	Latent class analysis, N=884	"Extreme-mixed" class – HIGH hyper- AND hypo- simultaneously – identified as distinct ASD subtype	24214165
Lane et al., 2010	Cluster analysis	Mixed profile (hyper in some modalities, hypo in others) was the MOST COMMON pattern	20093776
Tavassoli et al., 2014	Adults with ASD; Sensory Perception Quotient	Elevated both hyper- and hypo-sensitivity; "spiky" profile more extreme than controls	24327295

The DSM-5 acknowledges this with Criterion B: "Hyper- or hypo-reactivity to sensory input or unusual interest in sensory aspects of the environment" – but provides no mechanistic explanation for why both directions co-exist.

Evidence strength: [STRONG] (Strong – meta-analytic confirmation)

5.2 Component 1: TIMED Selection – Which Modalities Are Affected

Prediction: If OPC maturation arrest occurs during a specific gestational window, the sensory modalities most affected should correspond to the tracts that were myelinating during that window.

Gestational Week of Myelination	Tract	Sensory Modality	Predicted Effect if Insult at GW 23 – 32
20 – 28	Inferior cerebel-lar pedun-cle, medial lemnis-cus	**Proprioception, vestibular, deep touch**	**MOST affected** – squarely in vulnerability window

Gestational Week of Myelination	Tract	Sensory Modality	Predicted Effect if Insult at GW 23 – 32
24 – 32	Optic radiation, acoustic radiation	**Vision, hearing**	**Partially affected** – insult catches early myelination phase
28 – 36	Corticospinal tract, posterior limb IC	Motor control	Moderately affected (later part of window)
32 – 40+	Corpus callosum (splenium), optic radiation continuation	Cross-modal integration	Less affected (window closing)
Postnatal 0 – 6 mo	Corpus callosum (genu), frontal WM	Frontal executive, attention	Affected at demand spike, not directly by insult
Postnatal 6 – 24 mo	**Association fibres (AF, SLF, UF)**	**Language, social cognition**	**Affected at demand spike** – this produces the "growing into deficit" regression at 12 – 24 months

Does this match clinical observation?

Clinical Pattern in ASD	Timing Model Prediction	Match?
Proprioceptive hypo-sensitivity (poor body awareness, toe-walking, need for deep pressure)	Proprioceptive tracts myelinate GW 20 – 28 – squarely in window	**YES**
Vestibular differences (poor balance or vestibular seeking) are extremely common	Vestibular tracts myelinate GW 20 – 28	**YES**
Auditory hyper-sensitivity is one of the most commonly reported features	Auditory radiation begins ~GW 24 – 32 – insult catches early phase, producing partial deficit	**YES**
Visual processing shows mixed pattern (enhanced detail detection, impaired motion)	Optic radiation begins ~GW 24 – partial effects expected	**YES**
Language and social cognition emerge later (12 – 24 months) as defining features	AF, SLF, UF myelinate postnatally; hit demand spike at 12 – 24 months	**YES**

Evidence strength: [MODERATE] (Moderate – the individual elements are independently well-supported, but the specific correlation between gestational insult timing and modality-specific sensory profile has not been tested in a single cohort)

5.3 Component 2: TEMPORAL Expression – What Modulates Severity

Prediction: The same autistic individual should show fluctuating sensory sensitivity modulated by physiological state – because partially myelinated circuits have a reduced safety factor and become unreliable under physiological stress.

Modulating Factor	Evidence in ASD	Mechanism
Fatigue	Sensory over-responsivity worse after poor sleep (Mazurek & Petroski, 2015; PMID: 25735789); sleep disruption amplifies FULL sensory profile (Tudor et al., 2012; PMID: 22987127)	Metabolic depletion –> reduced ATP –> reduced safety factor –> conduction failure
Stress	Higher cortisol = greater sensory distress; lower cortisol = greater sensory seeking *in same child across days* (Bitsika et al., 2015; PMID: 25227209)	Cortisol modulates neural excitability and E/I balance
Autonomic state	Variable EDA responses to SAME stimuli across testing sessions (Schaaf et al., 2015; PMID: 25766683); sympathetic activation increases sensory aversion (Goodwin et al., 2006; PMID: 16803601)	Autonomic state determines sensory processing mode (Porges, 2007)
Environmental context	Same child shows different sensory responses at home vs school (Hochhauser & Engel-Yeger, 2010; PMID: 20015781)	Environmental demand load varies; partially myelinated circuits fail under higher environmental complexity
Hormonal cycle	Premenstrual worsening of sensory over-responsivity in autistic women (Pohl et al., 2019; PMID: 31820153)	Progesterone withdrawal reduces GABAergic inhibition –> reduced safety factor in partially myelinated inhibitory circuits
Time of day	Sensory over-responsivity worse in evening (Mazurek & Petroski, 2015)	Cumulative fatigue depletion across the day

Evidence strength: [MODERATE] (Moderate – documented across multiple independent studies, but no study has simultaneously tracked sensory thresholds with physiological state measures within the same individuals over time)

5.4 The Conduction Biophysics – Why Partial Myelination Produces Oscillation

The mechanism underlying temporal expression is not speculative. It is established neurophysiology from the multiple sclerosis literature:

5.4.1 Safety Factor Reduction In fully myelinated axons, the safety factor for conduction is 5 – 7 (Smith & McDonald, 1999; PMID: 10399936). This provides substantial margin – signals are transmitted reliably regardless of physiological state. In partially myelinated axons (thin sheaths, fewer wraps,

incomplete internodal coverage), the safety factor approaches 1.0:

- **Safety factor > 1.0**: Signal successfully transmitted
- **Safety factor ~ 1.0**: Conduction is probabilistic – sometimes works, sometimes doesn't
- **Safety factor < 1.0**: Conduction block – signal fails

Evidence: Waxman, 1977 (PMID: 328542); Felts et al., 1997 (PMID: 9153151); McDonald & Sears, 1970 (PMID: 5498340).

5.4.2 Frequency-Dependent Conduction Block Partially myelinated axons can transmit single impulses but fail at physiological frequencies. At 30 – 50 Hz (typical for sensory transmission), partially demyelinated axons showed ~50% conduction failure (McDonald, 1974; PMID: 4472101). This is critical because sensory processing requires sustained, high-frequency transmission during active perception.

5.4.3 Temperature-Dependent Conduction Failure Raising temperature by as little as 0.2 – 0.5 degreesC causes complete conduction block in partially myelinated axons (Rasminsky, 1973; PMID: 4710797; Davis & Jacobson, 1971; PMID: 4935068). This is reversible – cooling restores conduction (Guthrie & Nelson, 1977; PMID: 561977). The axon is intact; only the safety factor is marginal.

5.4.4 Stress-Induced Amplification Nitric oxide – released during inflammation and stress – causes reversible conduction block in partially myelinated axons at concentrations that do not affect normally myelinated axons (Redford et al., 1997; PMID: 9037398). This provides a direct mechanism: stress –> inflammation –> NO –> conduction block *selectively* in vulnerable circuits.

5.4.5 Two Failure Modes From One Mechanism Partial myelination predicts two qualitatively different failure modes, depending on which circuits fail:

Mode	Circuit Failure	Behavioural Presentation	When
Hyper-sensitivity	Inhibitory circuits fail (higher metabolic demand, longer axons, more myelination-dependent) –> excitatory dominance –> amplified signal	Overwhelming sensory experience; distress, avoidance, covering ears/eyes	Under moderate physiological stress – inhibitory circuits fail first
Hypo-sensitivity	Complete conduction block across both excitatory and inhibitory circuits –> signal dropout	Reduced perception; apparent indifference to sensory input	Under severe physiological stress/fatigue – all circuits fail

This resolves the paradox of co-occurring hyper- and hypo-sensitivity: they are not opposite conditions requiring different explanations. They are different failure modes of the same partially myelinated circuits, determined by current physiological load.

Evidence strength for the biophysical mechanism: [STRONG] (Strong – decades of MS research)
Evidence strength for its application to ASD: [SPECULATIVE] (Speculative – novel application; not yet tested in ASD populations)

5.5 Uhthoff's Phenomenon – The Direct Analogue

Uhthoff's phenomenon (Uhthoff, 1890) refers to the transient worsening of neurological symptoms in MS patients during physical exertion or elevated body temperature. The mechanism is precisely the safety-factor reduction described above: exercise-induced heating decreases the safety factor in partially myelinated circuits below the conduction threshold (Davis et al., 2010; PMID: 20536885; Frohman et al., 2013; PMID: 23401504). It is reversible upon cooling.

No published study has proposed an Uhthoff-like mechanism in autism. This paper proposes it as a direct analogue: if ASD involves partially myelinated circuits (due to developmental OPC arrest rather than autoimmune demyelination in MS), the same physiological modulation would produce the same fluctuating neurological function.

The Fever Paradox: The "fever effect" in autism (~80% of parents report behavioural improvement during fever; Curran et al., 2007; PMID: 17347881) appears to contradict this model. However, fever is not just temperature: it activates the locus coeruleus-noradrenergic (LC-NA) system (Mehler & Purpura, 2009; PMID: 19560900), which increases axonal excitability and raises the safety factor – potentially more than compensating for temperature-induced reduction. The model predicts that exercise-induced temperature increase (without LC-NA activation) should worsen sensory processing in ASD, while fever (with LC-NA activation) should improve it. This is a testable prediction.

Evidence strength: [SPECULATIVE] (Speculative – mechanistically coherent, novel, untested)

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6. The Gestational Fingerprint – Mapping ASD Symptoms to Specific Weeks

6.1 The Master Symptom-Timing Overlay

If each sensory modality and cognitive function depends on a specific white matter tract, and each tract myelinates on a defined schedule, then the specific symptom profile of an individual with ASD should act as a **forensic fingerprint** of when the gestational insult occurred.

Gestational Window of Insult	Tracts Most Affected	Predicted Dominant ASD Symptom Profile	Demand Spike Age
GW 20 – 24	Medial lemniscus, inferior cerebellar peduncle, vestibular pathways	**Severe proprioceptive/vestibular** dysfunction; motor delay; poor balance; sensory-seeking of deep pressure	6 – 18 months (motor milestones)
GW 24 – 28	Optic radiation (early), acoustic radiation (early), insular thalamocortical	**Auditory/visual hyper-sensitivity** + gustatory/interoceptive integration failure; severe feeding difficulties	12 – 24 months (so-cial/communicative demands)
GW 28 – 32	Corticospinal tract, corpus callosum (splenium), posterior IC	**Motor coordination** difficulties; cross-modal integration failure; DCD-like presentation	18 – 36 months (complex motor + language)
GW 32 – 36	Corpus callosum (continued), frontal WM (early)	**Milder motor** profile; executive difficulties; less sensory; more cognitive	5 – 7 years (academic demands)
Postnatal (birth – 6 mo)	CC genu, frontal WM	**Attention/executive** dominant (ADHD-like); mild sensory	5 – 12 years (sustained attention demands)

6.2 The Stacking Dimension – Duration Matters

A brief insult (hours to days) at one gestational age produces a narrow-window injury – one or two tracts affected, producing a "pure" sensory or cognitive phenotype. A sustained insult spanning multiple weeks (as in chronic preeclampsia) hits multiple tracts sequentially, producing the "full spectrum" ASD phenotype:

Insult Duration	Tracts Affected	Predicted Phenotype
Brief (hours) at GW 26	Acoustic radiation early phase	Isolated auditory hyper-sensitivity; may not meet ASD criteria
1 week at GW 26 – 27	Acoustic + optic radiation + insular	Auditory + visual + feeding; sub-threshold ASD features
4 weeks (GW 24 – 28)	**Multiple sensory + insular + vestibular**	**Classic ASD presentation** – multi-modal sensory, feeding, social interaction
8 weeks (GW 24 – 32)	**All sensory + motor + insular + beginnings of association**	**Severe ASD** – profound multi-system involvement

EOPE vs LOPE differential: As established in Carpenter (2026b), early-onset preeclampsia (EOPE, <34 wk) produces chronic HIF activation spanning much of the super-window, predicting severe multi-system ASD with feeding difficulties. Late-onset preeclampsia (LOPE, >=34 wk) produces subacute, oscillating HIF in the later phase, predicting milder sensory profiles with more cognitive/executive features and fewer feeding difficulties.

Evidence strength: [SPECULATIVE] (Speculative – the individual developmental timelines are established, but the specific mapping of insult week –> ASD symptom profile has not been tested)

6.3 The Oscillation Dimension – HIF Cycling and the Worst Outcomes

As established in Carpenter (2026b, 2026c), oscillating preeclampsia – where maternal blood pressure spikes and recovers repeatedly – produces both HIF-1a stabilisation (during spikes) and ROS-mediated oxidative damage (during recovery). This HIF cycling is worse than either sustained hypoxia or sustained normoxia because it activates both injury pathways alternately.

For the insular timing hypothesis, HIF cycling adds a specific prediction: **insular white matter connections subject to oscillating HIF would show MORE variable myelination than those subject to sustained HIF.** This would produce: - Greater circuit-to-circuit variability (some internodes myelinated, others not – even within the same tract) - Greater temporal dispersion of compound action potentials (Bostock & Sears, 1978; PMID: 702621) - **More extreme sensory oscillation** (wider swings between hyper- and hypo-sensitivity) - **More unpredictable "good days and bad days"**

The oscillation in maternal physiology produces oscillation in fetal myelination, which produces oscillation in postnatal sensory experience. The temporal pattern is not random – it is inherited from the gestational molecular environment.

Evidence strength: [SPECULATIVE] (Speculative – but mechanistically coherent with established HIF cycling evidence)

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7. Sensory Overload, Meltdown, and Shutdown – A Conduction Failure Model

7.1 The Three Stages

The overload –> meltdown –> shutdown progression reported by autistic individuals maps onto sequential conduction failure in partially myelinated circuits:

Stage	Circuit-Level Event	Behavioural Presentation	MS Parallel
Overload	Cumulative metabolic stress in partially myelinated circuits; ATP depleting, extracellular K+ accumulating; safety factor progressively decreasing	Increasing distress; covering ears/eyes; attempting to leave environment	Fatigue-dependent symptom worsening (Schwid et al., 2003; PMID: 12552036)
Meltdown	**Inhibitory circuits fail first** (PV+ interneurons: higher firing rate, longer axons, more metabolically demanding) –> loss of inhibition –> unchecked excitatory activity	Loss of behavioural regulation; crying, screaming, self-injury	E/I imbalance from preferential inhibitory failure (Selten et al., 2018; PMID: 29430164)
Shutdown	**Excitatory circuits fail** –> widespread conduction block –> sensory input ceases reaching cortex	Withdrawal, unresponsiveness, non-verbal; apparent dissociation	Uhthoff-like global conduction block (Davis et al., 2010)
Recovery	Metabolic restoration (ATP replenished, ionic gradients restored, safety factor rises above threshold)	Gradual return to baseline; facilitated by rest, quiet, dim lighting, cooling	Recovery from MS fatigue (Schwid et al., 2003) – identical timecourse

7.2 Recovery Time as a Measure of Myelination Status

Novel prediction: If overload –> meltdown –> shutdown represents sequential conduction failure, then: - More severe myelination deficit –> **longer recovery time** - Better myelination (naturally or through intervention) –> **faster recovery** - Recovery should be accelerated by interventions that reduce metabolic demand: rest, sleep, reduced sensory input, **cooling** - Recovery times should **decrease with age** as activity-dependent myelination partially compensates

This matches clinical observation precisely: autistic meltdown/shutdown recovery is facilitated by quiet rooms, dim lighting, cooling, and sleep. These interventions reduce neuronal metabolic demand, enabling faster ATP replenishment and ionic gradient restoration.

Evidence strength: [SPECULATIVE] (Speculative – but well-matched to clinical observation and MS fatigue recovery literature)

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8. The Developmental Trajectory – From Birth to Adulthood

8.1 Age-Related Changes in Sensory Profiles

The developmental trajectory of sensory features in ASD maps onto the postnatal myelination schedule:

Age	Myelination Event	Predicted Sensory Pattern	Observed Pattern	Evidence
0 – 6 mo	Primary sensory/motor tracts myelinating	Subtle differences, masked by immature baseline	Detectable by 6 – 12 mo prospectively	Ozonoff et al., 2010
6 – 24 mo	Rapid myelination of CC, IC, optic/acoustic radiations	Tracts hit demand spike; if OPC-arrested, fail to meet demand –> **hypo-responsivity dominant**	Hypo-responsivity most prominent	Baranek et al., 2013 (PMID: 23695716)
2 – 6 yr	Association fibres; inhibitory circuits maturing	Compensatory excitability –> E/I imbalance –> **hyper-responsivity emerging**	Hyper-responsivity peaks ages 6 – 9	Ben-Sasson et al., 2009 (PMID: 18615687)

Age	Myelination Event	Predicted Sensory Pattern	Observed Pattern	Evidence
6 – 12 yr	Continued association fibre maturation; experience-dependent myelination	Partial compensation through activity-dependent myelination	Some sensory normalisation	Kern et al., 2006 (PMID: 16897387)
Adolesc	Frontal WM; cingulum maturing	Better top-down control; hormonal disruption of marginal circuits	Continued improvement + puberty-related instability	McCormick et al., 2016 (PMID: 26861713)
Adultho	Cingulum peaks (FA >40 yr)	Full prefrontal gating available	Never fully normalises but improves	Kern et al., 2006

8.2 The Activity-Dependent Myelination Rescue

A crucial feature of the OPC model is that activity-dependent myelination (Fields, 2015; PMID: 25936579) provides a partial rescue mechanism. Neuronal activity drives OPC proliferation and differentiation – meaning intensive therapy is not just "practice" but a biological intervention that stimulates myelination.

This explains three otherwise puzzling clinical observations:

1. **Why therapy works but plateaus**: Activity-dependent myelination gradually thickens sheaths –> but the OPC pool is depleted –> improvement continues until available OPCs are exhausted –> plateau
2. **Why earlier intervention is better**: The "forming" phase (active myelination) responds more to activity-dependent stimulation than the "formed" phase (OPC pool depleted)
3. **Why improvement never reaches fully typical levels**: The starting point was thinner myelin from a depleted OPC pool; activity-dependent addition brings closer to normal but never achieves full thickness

Evidence strength: [MODERATE] (Moderate – activity-dependent myelination is well-established; its specific application to ASD therapy response is indirect)

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9. Feeding Difficulties – The Insular Cortex as Explanatory Hub

9.1 Reframing ASD Feeding Difficulties

The insular timing hypothesis reframes each category of ASD feeding difficulty as a specific failure mode of insular cortex multi-modal integration:

Feeding Difficulty Category	Traditional Explanation	Insular Integration Explanation
Food selectivity / picky eating (~70%)	Behavioural rigidity; restricted interests	Multi-modal integration failure – taste + texture + temperature + smell cannot be bound into a coherent percept; child restricts to the few foods that produce predictable, tolerable sensory signatures
Food refusal (~40 – 60%)	Oppositional behaviour; control	The gustatory-interoceptive integration pathway (intrainsular tract: posterior –> anterior insula) is undermyelinated – food triggers fragmented, aversive, or absent-of-meaning signals rather than a unified "food" percept

Feeding Difficulty Category	Traditional Explanation	Insular Integration Explanation
Texture selectivity – the most distinctive ASD feeding feature	Sensory sensitivity	Somatosensory input from oral cavity (posterior insula) fails to integrate with gustatory input (anterior insula) – a specific intrainsular tract conduction failure
Eating only brands/colours/shapes	Cognitive rigidity	Visual features (processed pre-insularly) provide predictability that compensates for unreliable gustatory/interoceptive integration – a rational compensatory strategy, not pathological rigidity
Mealtime behavioural rigidity (~50 – 70%)	Executive function deficit	Reducing environmental variability reduces total sensory processing load, keeping partially myelinated insular circuits below their failure threshold – an adaptive strategy for maintaining circuit function

9.2 Pre-Diagnostic Feeding Trajectory

The Ashley et al. (2020; PMID: 31800528) prospective infant-sibling study provides critical evidence:

- Feeding difficulties **measurably diverge from 15 months** in children later diagnosed with ASD
- The trajectory steepens between 15 – 36 months
- This timeline aligns with the postnatal myelination demand spike for insular connections
- By 36 months, ASD children have significantly higher total feeding difficulty scores than all comparison groups

Key implication: Feeding difficulty trajectory may be a pre-diagnostic biomarker for ASD – and the insular timing hypothesis explains why. The insular white matter connections required for food integration hit their demand spike at exactly the age when feeding difficulties diverge.

9.3 Overlap with Preterm Feeding Difficulties

Preterm birth and ASD share convergent feeding difficulty features:

Feature	Preterm	ASD	Shared Mechanism (Insular Timing)
Oral motor difficulties	Immature coordination, weak suck	Oral motor delays	Both: insular cortex opercularization disrupted during same GW window
Texture sensitivity	Oral sensory defensiveness	Primary driver of food selectivity	Both: intrainsular tract undermyelinated –> somatosensory/gustatory integration failure
Food refusal	Especially during solid transition	High rates	Both: multi-modal integration failure at the gustatory cortex
GI comorbidity	High prevalence	Elevated (Micai et al., 2023; PMID: 37913872)	Both: insular autonomic regulation of gut motility impaired

This convergence is predicted by the model: preterm birth IS the clinical setting where OPC maturation

arrest was first characterised (Back et al., 2001). Preterm-born children and children with ASD share insular/white matter developmental vulnerability because they share the underlying mechanism – OPC injury during the super-window.

Evidence strength: [MODERATE] (Moderate – convergent evidence from two literatures, but no direct comparison study exists)

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10. Treatment Convergence – Why Empirical Therapies Work

10.1 The Treatment Puzzle

Multiple feeding and sensory interventions demonstrate efficacy in ASD, but most lack an established mechanism. The insular timing hypothesis provides a unifying framework by identifying six target mechanisms:

- **(a)** Taste receptor function (peripheral)
- **(b)** Insular cortex processing (central)
- **(c)** White matter connectivity / myelination support
- **(d)** Interoceptive regulation
- **(e)** Autonomic / vagal tone
- **(f)** OPC / myelination direct support

10.2 Treatment – Mechanism Mapping

Treatment	Evidence Level	Proposed Mechanism(s)	How Insular Timing Hypothesis Explains Efficacy
D-Cycloserine + exposure	RCT: Doubled feeding intervention efficacy (76% vs 37%; Sharp et al., 2017; PMID: 28632204)	(b) NMDA partial agonist –> glutamatergic facilitation of insular cortical plasticity	D-cycloserine enhances extinction learning at the insular cortex (fear/disgust extinction) AND may promote activity-dependent myelination through enhanced NMDA-mediated neuronal firing
Zinc supple-mentation	Multinational reanalysis confirms zinc deficiency as reproducible biochemical feature of ASD (2026; PMID: 41372683)	(a) + (c) Zinc is critical for taste bud turnover (every 10 – 14 days) AND for Shank3 synaptic scaffolding	Zinc deficiency compounds cortical integration failure with peripheral taste receptor dysfunction – restoring zinc addresses both the peripheral taste signal AND central synaptic function

Treatment	Evidence Level	Proposed Mechanism(s)	How Insular Timing Hypothesis Explains Efficacy
DHA / Omega-3	MFSD2A transporter mutations cause hypomyelination (Nguyen et al., 2014; PMID: 30043326); n-3 PUFAs protect oligodendrocytes (Elbaz & Bhatt, 2017; PMID: 27714669)	(c) + (f) DHA is structurally required for myelin membrane synthesis; n-3 PUFAs directly protect OPCs	Provides the substrate for activity-dependent myelination to proceed – partially myelinated insular tracts cannot thicken without adequate DHA supply
Sensory integration therapy (Ayres-based)	Moderate evidence from systematic reviews	(b) + (c) Structured multi-sensory input drives activity-dependent myelination of insular integration circuits	Repetitive, graded sensory input specifically targets the circuits that are undermyelinated – providing the neuronal activity that drives OPC differentiation
Weighted blankets / deep pressure	Cross-domain effects including mealtime (Bestbier & Williams, 2017; PMID: 34184958)	(d) + (e) Deep pressure activates C-tactile afferents –> posterior insula –> autonomic calming via vagal tone modulation	Reduces autonomic arousal, which RAISES the safety factor for conduction in partially myelinated circuits – bringing them above the conduction threshold during meals
Oral motor therapy / desensitisation	Clinical evidence; limited RCT	(a) + (b) Peripheral receptor habituation + cortical plasticity	Graded oral sensory input drives activity-dependent myelination of the specific intrainsular tract (posterior –> anterior insula) responsible for oral-gustatory integration
Food chaining	Clinical evidence	(b) Systematic expansion of predictable sensory signatures	Builds cortical representations incrementally, reducing the integration load on partially myelinated circuits by starting with near-identical sensory patterns and gradually expanding
GFCF diet	Mixed RCT evidence; parental reports of improvement	(d) Possible reduction of gut-mediated interoceptive noise via reduced inflammatory signalling	If gluten/casein metabolites increase gut inflammation –> insular interoceptive processing load increases –> undermyelinated insular circuits pushed closer to failure threshold; removing them reduces interoceptive “noise”
Probiotics / gut-brain axis	Emerging RCT evidence	(d) + (e) Gut microbiome –> vagal afferents –> insular interoceptive processing chain	Modulates the interoceptive signal stream arriving at the posterior insula – reducing processing demand on undermyelinated circuits

Treatment	Evidence Level	Proposed Mechanism(s)	How Insular Timing Hypothesis Explains Efficacy
Melatonin	Neuroprotective; myelination-supporting (Kumar et al., 2007; PMID: 17296249); improves sleep –> reduces daytime sensory over-responsivity (Tordjman et al., 2012; PMID: 22963932)	(c) + (f) Direct myelinotrophic effects + improved sleep –> improved circuit recovery overnight	Addresses the well-documented melatonin deficiency in ASD that compounds existing myelination deficit; better sleep –> better overnight circuit recovery –> higher safety factor the next day

10.3 Why D-Cycloserine Is the Strongest Validation

The Sharp et al. (2017) finding deserves emphasis: D-cycloserine (an NMDA partial agonist originally used for tuberculosis, then anxiety) doubled the efficacy of feeding intervention in ASD. No purely behavioural model explains this. The insular timing hypothesis does:

1. NMDA receptor activation drives long-term potentiation at cortical synapses
2. Enhanced synaptic activity drives OPC proliferation and differentiation (Gibson et al., 2014)
3. D-cycloserine therefore amplifies the activity-dependent myelination signal at insular circuits *during* feeding exposure therapy
4. The result: circuits that were conducting unreliably are pushed toward better myelination by pharmacologically enhanced neuronal firing during therapeutic feeding exposure

This reframes D-cycloserine not as an anxiolytic that happens to help feeding, but as a myelination accelerant that works by enhancing the activity-dependent myelination signal at the insular cortex during structured sensory exposure.

Evidence strength: [MODERATE] (Moderate – the RCT result is strong; the mechanistic interpretation via insular myelination is novel)

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11. Autobiographical Sensory Memory – An Unexplored Dimension

11.1 The Memory Gap

No published study has specifically examined sensory autobiographical memory in autistic individuals – whether they can recall childhood taste, smell, or texture experiences. This represents a significant literature gap.

What is known:

Finding	Evidence	PMID
ASD shows impaired episodic autobiographical memory (AM) but preserved semantic AM	Multiple studies	17668308; 33954915; 32191044
Sensory encoding appears intact; the deficit is in *binding* sensory fragments into coherent narratives	Gaigg & Bowler, 2012	32869928

Finding	Evidence	PMID
Visual/sensory cues compensate for narrative AM deficits – showing a picture helps retrieve memories that spontaneous recall cannot access	Crane et al., 2012	31354565
A 2025 study demonstrates direct **insula –> hippocampus connections** for memory encoding	Cabral-Calderin et al., 2025	40659846

11.2 Prediction from the Insular Timing Hypothesis

If the intrainsular tract and insular-hippocampal connections are partially myelinated:

1. **Sensory encoding** (at primary sensory cortices) would be **intact or enhanced** – detail is captured
2. **Sensory-to-narrative binding** (requiring insular integration + hippocampal encoding) would be **impaired** – the experience is recorded as fragments, not as a coherent story
3. **Cue-dependent retrieval** should be **superior with sensory cues** (bypassing the integration pathway) compared to verbal/narrative prompts

This is consistent with Temple Grandin's extensive first-person accounts: vivid sensory memory of childhood experiences (enhanced sensory fidelity) combined with difficulty organising these into narrative autobiographical sequences (impaired integration).

Evidence strength: [SPECULATIVE] (Speculative – literature gap identified; prediction made; not yet tested)

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12. Adversarial Stress Test

12.1 What This Model Claims

The Insular Timing Hypothesis claims that: 1. ASD feeding difficulties arise from OPC maturation arrest affecting insular cortex white matter connections during the gestational super-window 2. The specific sensory modalities affected are determined by insult timing (TIMED selection) 3. Sensory oscillation between hyper- and hypo-sensitivity is produced by physiological state modulation of conduction reliability in partially myelinated circuits (TEMPORAL expression) 4. The overload –> meltdown –> shutdown sequence represents sequential conduction failure 5. Empirical feeding/sensory treatments work by targeting taste receptor function, insular processing, myelination support, or autonomic regulation of circuit operating conditions

12.2 What Would Disprove It

1. **No insular white matter abnormality in ASD:** If DTI studies consistently show NORMAL insular white matter integrity in ASD, the structural substrate fails. *Status: Contradicted – Failla et al. (2017), Aoki et al. (2013), Kilroy et al. (2022) all show insular region WM abnormalities.*
2. **No correlation between insular WM integrity and feeding/sensory measures:** If insular tract integrity does not predict sensory processing or feeding difficulty severity, the mechanistic link fails. *Status: Partially supported – Failla et al. (2017) demonstrates the intrainsular-tactile correlation.*
3. **Sensory profiles not modality-specific:** If ASD sensory profiles are UNIFORM across modalities (equally hyper or hypo in all senses), the TIMED component fails. *Status: Contradicted – modality-specific "spiky" profiles are the dominant pattern (Lane et al., 2010; Tavassoli et al., 2014).*
4. **Sensory oscillation not state-dependent:** If sensory sensitivity is STABLE regardless of fatigue,

stress, sleep, or hormonal state, the TEMPORAL component fails. *Status: Contradicted – state-dependent oscillation documented (Kern et al., 2006; Schaaf et al., 2015; Bitsika et al., 2015).*

5. **Preterm-born children do NOT show elevated feeding difficulties:** If preterm birth (the clinical setting of confirmed OPC arrest) does not produce elevated feeding difficulties, the causal chain breaks. *Status: Contradicted – up to 40 – 70% of preterm infants have feeding difficulties.*
6. **Feeding difficulties in ASD are unrelated to sensory processing:** If feeding difficulties occur independently of broader sensory profiles, the insular convergence fails. *Status: Contradicted – feeding-sensory relationship confirmed by systematic review (Nimbley et al., 2022; PMID: 35737818).*

12.3 What Has NOT Been Tested

1. **No study has correlated gestational insult timing with modality-specific sensory profile** – the critical TIMED prediction
2. **No study has tested Uhthoff-like exercise challenge in ASD** – the critical TEMPORAL prediction
3. **No study has simultaneously measured DTI tract integrity and modality-specific sensory thresholds within the same ASD individuals**
4. **No longitudinal study has tracked insular WM development alongside feeding difficulty onset**
5. **No study has tested D-cycloserine's mechanism via myelination rather than anxiety extinction**

12.4 Honest Limitations

1. **Syndromic autism:** The model applies most naturally to ASD associated with preterm birth, perinatal complications, or white matter involvement. Purely genetic/synaptic autism (Fragile X, Rett, TSC) may produce sensory oscillation through E/I imbalance at the synaptic level rather than through myelination failure – though the two mechanisms are not mutually exclusive.
2. **DTI as proxy:** Fractional anisotropy reflects multiple microstructural properties, not myelination alone. Apparent "normal" DTI does not exclude functionally marginal myelination.
3. **Multifactorial feeding difficulties:** Not all ASD feeding difficulties are sensory-driven. Gastrointestinal comorbidity, motor coordination deficits, and anxiety all contribute independently. The insular timing hypothesis does not claim to explain 100% of ASD feeding problems – only the sensory-integration-driven component.
4. **Reverse causation risk:** It is possible that insular WM abnormalities in ASD are a consequence (not cause) of abnormal sensory experience – reduced sensory exploration –> less activity-dependent myelination –> thinner insular tracts. This is mitigated but not excluded by the prospective infant-sibling evidence (Ashley et al., 2020) showing feeding divergence from 15 months.

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13. Testable Predictions

#	Prediction	Study Design	Falsification Criterion
1	**DTI-sensory correlation:** Tract-specific myelin integrity should predict modality-specific sensory thresholds within ASD individuals – thin auditory radiation myelin –> auditory hyper-sensitivity; thin medial lemniscus –> proprioceptive differences	Cross-sectional: DTI + comprehensive psychophysical sensory battery in 50+ ASD adults	No significant tract-specific correlations
2	**Uhthoff-like challenge:** Exercise-induced body temperature increase should transiently worsen sensory processing in ASD; cooling should restore it. Fever should NOT worsen (due to LC-NA compensation)	Within-subject: temperature manipulation (exercise vs passive heating vs fever observation) + sensory thresholds pre/post	No temperature-dependent sensory change; or fever worsens equally
3	**Gestational timing correlation:** Children born preterm at 24 wk vs 28 wk vs 32 wk who develop ASD should have DIFFERENT sensory profiles corresponding to which tracts were myelinating at each age	Preterm-ASD cohort with detailed gestational records + modality-specific sensory profiling	Same sensory profile regardless of gestational age at insult
4	**Myelination-recovery correlation:** ASD individuals who show greater improvement in sensory features over time should show greater increase in insular white matter integrity (FA) over the same period	Longitudinal DTI + sensory measures at 2, 5, 10, 15 years in ASD cohort	No correlation between DTI change and sensory improvement

#	Prediction	Study Design	Falsification Criterion
5	**D-cycloserine mechanism:** D-cycloserine + feeding therapy should produce measurable increases in insular tract myelination (DTI) compared to feeding therapy alone – beyond what anxiety reduction alone would predict	RCT: D-cycloserine + therapy vs placebo + therapy, with DTI pre/post	No DTI difference; or behavioural improvement uncorrelated with DTI change

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14. Clinical Implications

14.1 Reframing Feeding Difficulties

If the insular timing hypothesis is correct, feeding difficulties in ASD should be recognised as **neurological** – not behavioural. This changes clinical approach:

- **Assessment:** Include sensory profiling (modality-specific) alongside feeding assessment, looking for the insular integration pattern (taste + texture + interoceptive + emotional binding failure)
- **Intervention:** Target the mechanism – activity-dependent myelination through graded, predictable, multi-sensory feeding exposure – rather than compliance
- **Pharmacological support:** Consider D-cycloserine, zinc, and DHA as mechanism-targeted adjuncts to feeding therapy, supported by the treatment convergence evidence
- **Environmental modification:** Reduce total sensory load during mealtimes (quiet, predictable environment) to keep partially myelinated insular circuits below their failure threshold

14.2 The Symptom Fingerprint as Diagnostic Tool

If gestational timing determines the symptom profile, then the **reverse** is also true: a detailed sensory/feeding/motor profile could estimate when the gestational insult occurred. This has implications for:

- **Pregnancy surveillance:** If early feeding difficulties in a preterm-born infant suggest insular involvement consistent with a GW 24 – 28 insult, screen for other systems vulnerable at that window (renal, cardiac, cerebellar)
- **Anticipatory guidance:** Parents of preterm-born infants could receive evidence-based expectations for which sensory modalities are most likely to be affected based on gestational age at delivery
- **Research stratification:** ASD cohorts could be stratified by gestational history to reduce heterogeneity – a major methodological challenge in ASD research

14.3 From Symptom Management to Mechanism-Targeted Therapy

The treatment convergence table (Section 10) demonstrates that apparently unrelated feeding interventions all target the same mechanisms. This suggests a new intervention framework:

1. **Support peripheral taste function** (zinc supplementation – addressing a reproducible biochemical deficit)
2. **Provide myelination substrate** (DHA/omega-3 – structural requirement for myelin membrane)
3. **Drive activity-dependent myelination** (structured sensory exposure + optional D-cycloserine enhancement)
4. **Reduce circuit operating stress** (autonomic regulation: deep pressure, vagal tone, melatonin for

sleep)
5. **Monitor progress via DTI** (insular tract integrity as an objective outcome measure)

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15. Significance Statement – What This Paper Adds

This paper adds the insular cortex – previously absent from the gestational oxygen-timing alignment model – as an eleventh organ system with its own vulnerability window, progenitor population, and latent failure mechanism. More broadly, it:

1. **Connects feeding difficulties to a gestational mechanism** – for the first time providing a specific neuroanatomical and developmental explanation for why 46 – 89% of autistic children struggle with food
2. **Introduces the TIMED + TEMPORAL dual model** – resolving the paradox of co-occurring hyper- and hypo-sensitivity through established conduction biophysics
3. **Applies Uhthoff's phenomenon to autism** – a novel cross-disciplinary translation from MS that explains sensory oscillation, state-dependent worsening, and the meltdown –> shutdown sequence
4. **Maps ASD symptoms to gestational weeks** – creating a forensic fingerprint framework that transforms the heterogeneous ASD phenotype into a timing-specific outcome
5. **Explains why empirical treatments work** – ten apparently unrelated feeding/sensory interventions converge on the same six mechanisms predicted by the insular timing hypothesis
6. **Identifies the most accessible entry point** for the oxygen-timing alignment model – feeding difficulties are visible, relatable, experienced by almost every ASD family, and surrounded by successful empirical treatments awaiting mechanistic explanation

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Declarations

Conflict of interest: None declared. **Funding:** None – independent research. **Data sharing:** All evidence cited is from publicly available PubMed-indexed literature. No original data were collected. **Ethics:** Not applicable – synthesis of published evidence. **Author note:** The author is the parent of a child initially diagnosed with ASD whose feeding and sensory difficulties motivated this investigation. This personal connection is declared as a potential source of confirmation bias. All claims are graded for evidence strength and the adversarial stress test (Section 12) is included to facilitate critical assessment.

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